

Effect of Plastic Compressibility on Dynamic Crack Growth

Alan Needleman

Department of Materials Science and Engineering Texas A&M University, College Station, TX 77843

Dynamic crack growth is analyzed numerically for a plane strain block with an initial central crack and subject to impact loading. The material is characterized as a plastically compressible isotropically hardening elastic-viscoplastic solid. A cohesive constitutive relation, that relates tractions and displacement jumps, is also specified across the crack plane. The cohesive relation introduces a characteristic length into the formulation. Also, crack growth, when it occurs, emerges as a natural outcome of the loading history. Full transient analyses are carried out. Results are presented for the effects of plastic compressibility on the crack growth resistance, on the crack speed and on the evolution of crack tip fields.